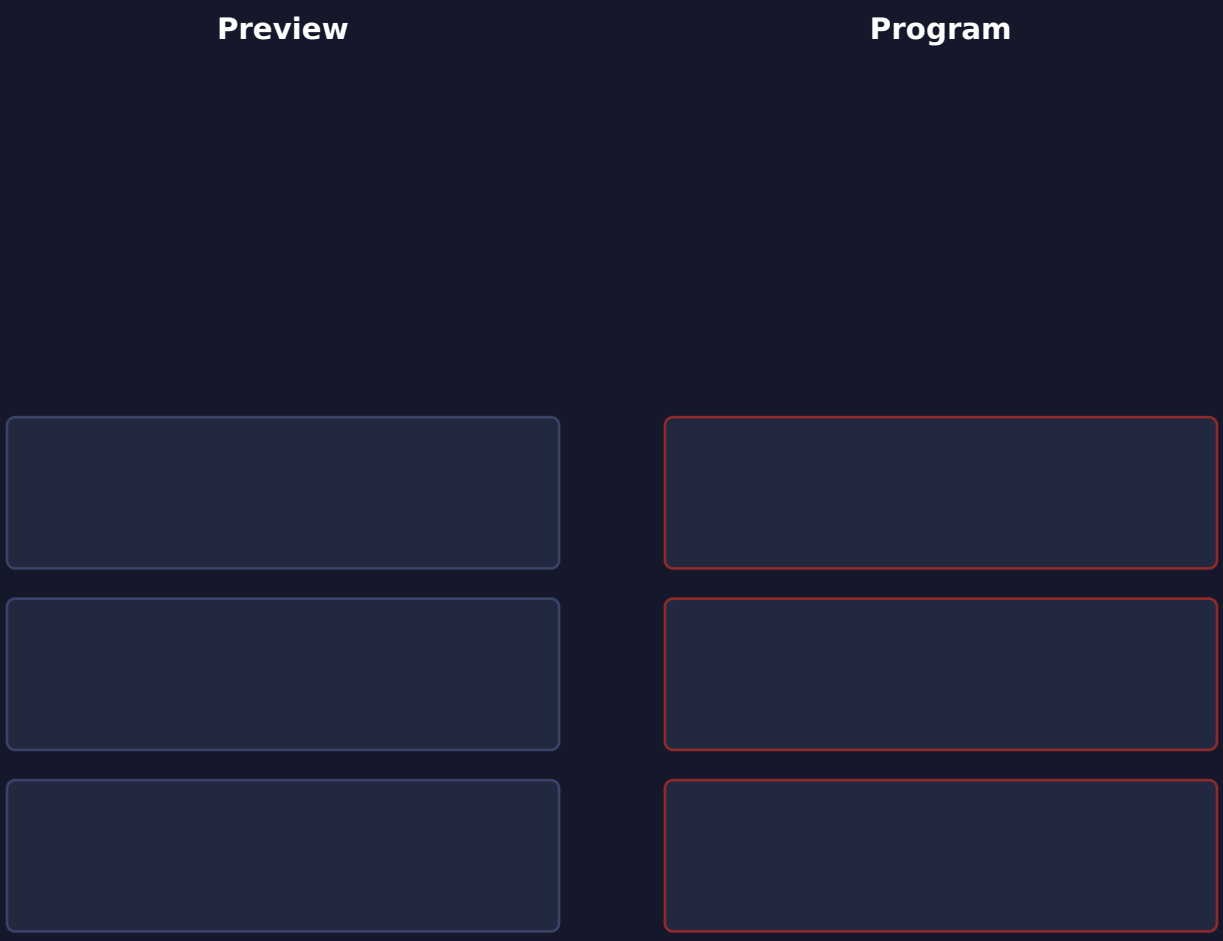


Multidestination Studio

User Guide · Version 1.0.3

A plugin for OBS Studio · Windows, macOS, Linux

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1. What is Multidestination Studio?

At a conference, a concert or a church service, the video team rarely sends the same picture everywhere. The LED wall behind the stage, a pair of projectors on the side screens, the confidence monitor for the speaker and the live stream all need different content, at different resolutions, switched at the same moment.

Multidestination Studio is a free plugin for OBS Studio that does exactly this. It adds independent **destinations** to OBS, each with its own canvas, resolution, Preview and Program, and lets you run all of them from one full-screen studio view with a single **Transition** or **Cut**. You keep everything you already know from OBS: sources, filters, transitions, audio, streaming and recording.

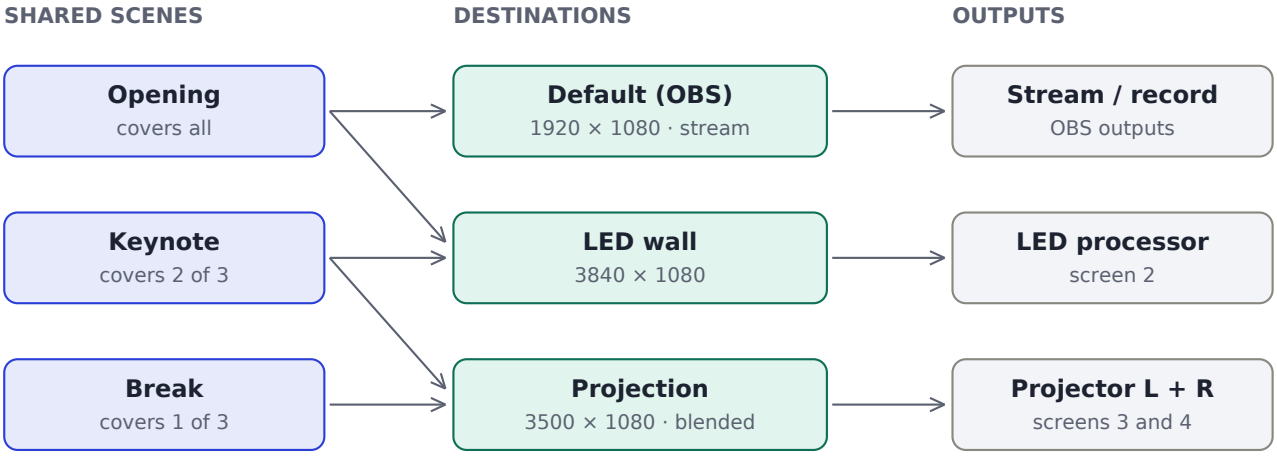
What it gives you

- **Several outputs, one operator.** Every destination has its own picture, but one scene list and one TAKE drive them all.
- **A different layout per screen.** The same scene can show a full-screen camera on the stream, a wide graphic on the LED wall and nothing at all on the projectors.
- **Projector blending built in.** Spread one destination across several projectors with soft-edge blending, gamma, black level and keystone correction.
- **A safe live output.** Changes you prepare in Preview never touch what is on air, unless you decide to edit the Program directly.

Requirements

Item	Requirement
OBS Studio	Version 31.1 or later
Systems	Windows 10/11 (64-bit), macOS (Apple Silicon and Intel), Ubuntu 24.04 (x86_64)
Hardware	One graphics card able to render every destination at your OBS frame rate. Each extra screen used for output must be connected to the computer.
Languages	English, French, Norwegian (Bokmål)

2. Key concepts



Each scene covers some destinations. Each destination keeps its own layout of the scene and feeds its own outputs.

Concept	Meaning
Destination	An independent video output with its own canvas and resolution: a screen, an LED wall, a group of blended projectors. The **default destination** is the OBS main canvas, the one OBS streams and records.
Scene	Scenes are shared. A scene lists the destinations it **covers**; each covered destination has its own version of the scene, with its own sources and layout.
Preview / Program	As in the OBS Studio Mode: Preview shows what is coming next, Program shows what is live. Each destination has both.
TAKE	**Transition** or **Cut** sends the Preview scene to Program on every destination it covers, at the same moment, using the OBS transition.
Lock	A locked destination ignores Transition and Cut and keeps its current Program.
Modify PGM	Program shows a copy of the scene taken live. Modify PGM lets you edit that live picture directly; the changes are also kept in the scene.
Output blending	A destination split across several outputs (usually projectors) whose overlaps are softly blended to look like one seamless image.

Good to know. A destination that a scene does not cover is not affected when that scene is taken: it simply keeps showing its current Program. This lets you change one screen without touching the others.

3. Installation and first launch

Close OBS, then install the package for your system, downloaded from the project's GitHub Releases page.

System	Package	Notes
Windows	multidestination-studio-1.0.0-windows-x64-installer.exe	The installer is not code-signed: Windows SmartScreen or Defender may ask you to confirm.
macOS	Universal .pkg package	Not signed: open System Settings → Privacy & Security → Open Anyway.
Ubuntu	multidestination-studio-1.0.0-x86_64-linux-gnu.deb	sudo apt install ./ followed by the file name.

Start OBS. A **Multidestination Studio** button appears in the OBS **Controls** dock, next to **Studio Mode**. The same command is in the **Tools** menu, and you can assign a hotkey to it in OBS Settings → Hotkeys.

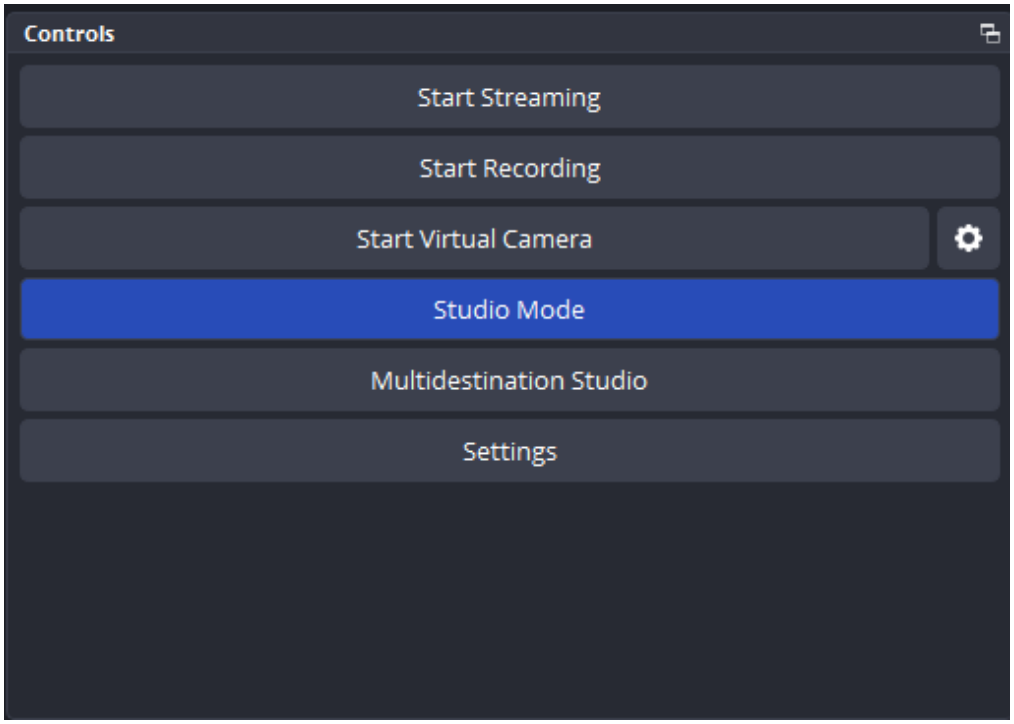


Figure 1. The Multidestination Studio button in the OBS Controls dock.

Multidestination Studio opens full screen. When it opens, OBS Studio Mode is switched on (and switched off again when you leave, if it was off), so that the default destination also has a real Preview and Program. To go back to the normal OBS window, use **File → Exit Multidestination Studio**, the **Quit** button in the top-right corner, or the **Exit** button in the Controls dock.

> **Tip.** To have the studio open automatically every time OBS starts, turn on **Configuration Settings → Startup Settings → Launch Multidestination Studio on startup**.

4. A tour of the studio

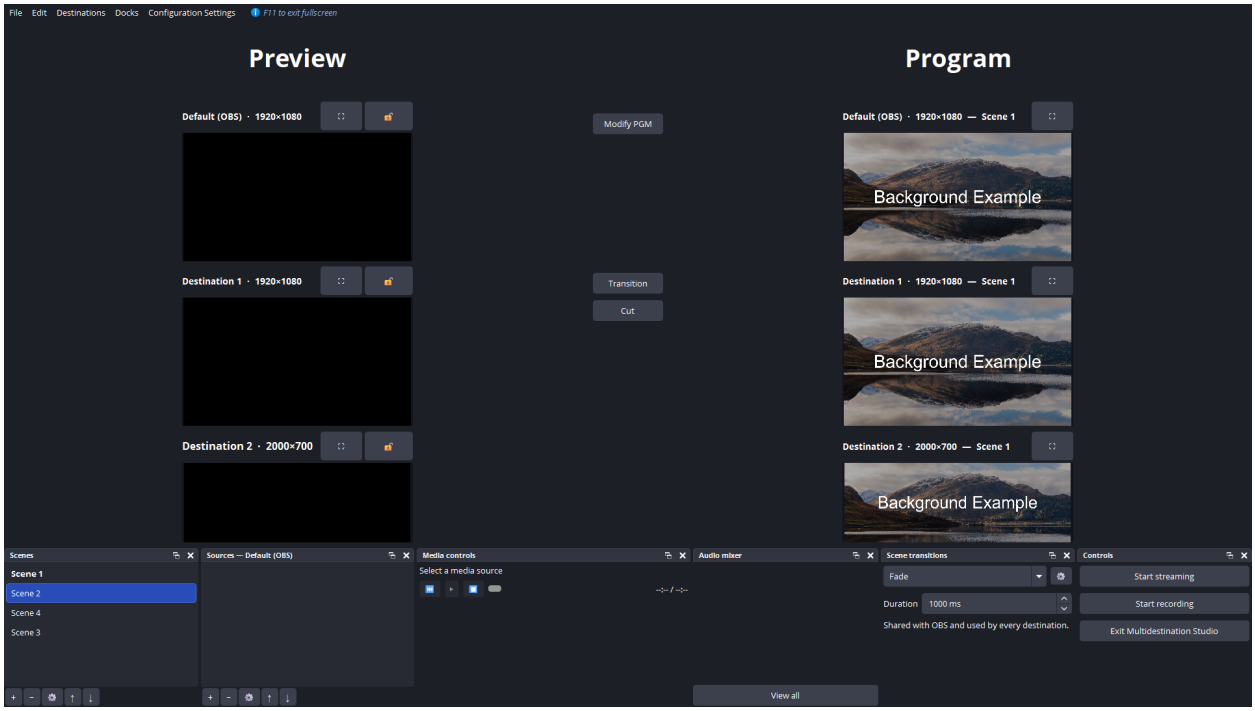


Figure 2. The Multidestination Studio window with three destinations.

Menus

Menu	Contents
File	Exit Multidestination Studio.
Edit	Undo, copy and paste, remove, transform, order, properties and filters for the selected source (see chapter 7).
Destinations	One submenu per destination, Manage destinations, Output blending, New destination (chapters 5 and 9).
Docks	Show or hide each dock, Reset docks, Fullscreen (F11).
Configuration Settings	After a transition, Destinations shown, Startup Settings, Language (chapter 11).

Next to the menus, an information hint reminds you that **F11** switches full screen on or off. In full screen, moving the mouse to the top-right corner reveals **Minimize**, **Exit fullscreen** and **Quit**; they disappear when the mouse moves away.

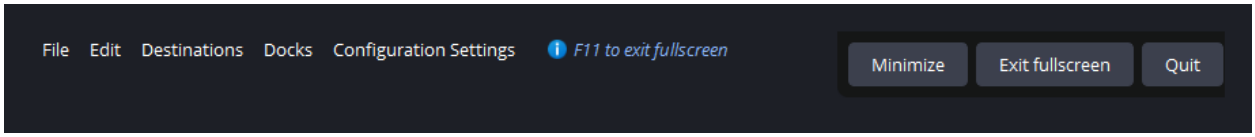


Figure 3. Top-right window controls in full screen, and the F11 hint next to the menus.

Preview and Program

The **Preview** column (left) and the **Program** column (right) show one tile per destination, stacked and scaled to fit the window. Each tile has a header with the destination name and resolution (in Program, also the scene on air) and two buttons:

- **Enlarge** shows only this destination, much larger, for precise placement. **Show all** brings the other destinations back.
- **Lock** (Preview only) keeps this destination out of Transition and Cut.

When space is short, the buttons shrink to their icons and the name text scales down so it always remains complete. Between the columns are **Modify PGM**, then **Transition** and **Cut**.



Figure 4. A tile header: destination name, Enlarge and Lock (unlocked and locked).

Docks

By default the six docks sit side by side at the bottom of the window: **Scenes**, **Sources**, **Media controls**, **Audio mixer**, **Scene transitions** and **Controls**. They can be moved, resized, floated or closed like OBS docks; the arrangement is saved with your scene collection. **Docks → Reset docks** restores the default layout.

Dock	Role
Scenes	The shared scene list: add, remove, rename, choose destinations, reorder.
Sources	Sources of the active tile (the destination you clicked last): add, remove, show or hide, reorder, properties.
Media controls	Play / pause, restart and stop the selected media source.
Audio mixer	Meters, faders, mute and monitoring for the active audio sources.
Scene transitions	The OBS transition and its duration, shared by every destination.
Controls	Start or stop streaming and recording, exit the studio.

5. Destinations

Creating a destination

1. Choose **Destinations → New destination…**.
2. Enter a **name** and a **resolution**. Choosing an **output screen** fills in the resolution of that screen automatically.
3. Tick **Add to existing scenes** if the new destination should appear in every scene you already have. Leave it unticked to add it to scenes one by one.
4. Click **Create**. The destination appears in Preview and Program for every scene that covers it.

When **All destinations, all the time** is on (chapter 11), a note under the checkbox reminds you that the new destination will be visible under every scene; where it was not added, it shows **Not in this scene** and nothing is placed on it.

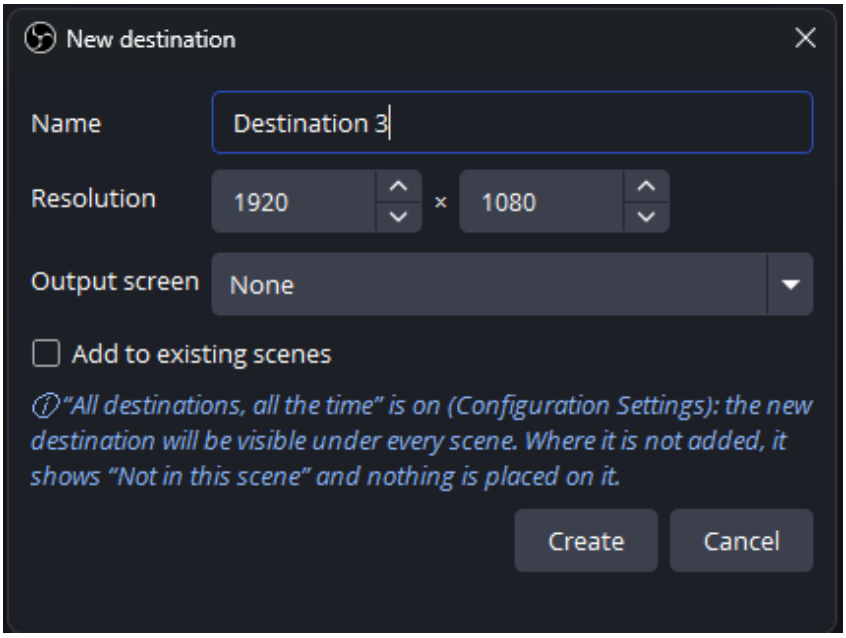


Figure 5. The New destination dialog, with the note shown when All destinations, all the time is on.

The Destinations menu

Each destination has its own submenu:

Command	What it does
Settings…	Rename, change the resolution or the output screen.
Output blending…	Split this destination across several projectors (chapter 9).
Open fullscreen on …	Shows the Program of the destination full screen on its output screen. For a blended destination: one full-screen window per projector.
Open windowed	Shows the Program in a normal window, for checking or for capture.
Close projection	Closes the full-screen or windowed output.
Delete	Deletes the destination and its layouts in every scene, after confirmation.

In a projection window, **Esc** closes the projection and a double-click switches between full screen and window. The mouse pointer is hidden on output screens.

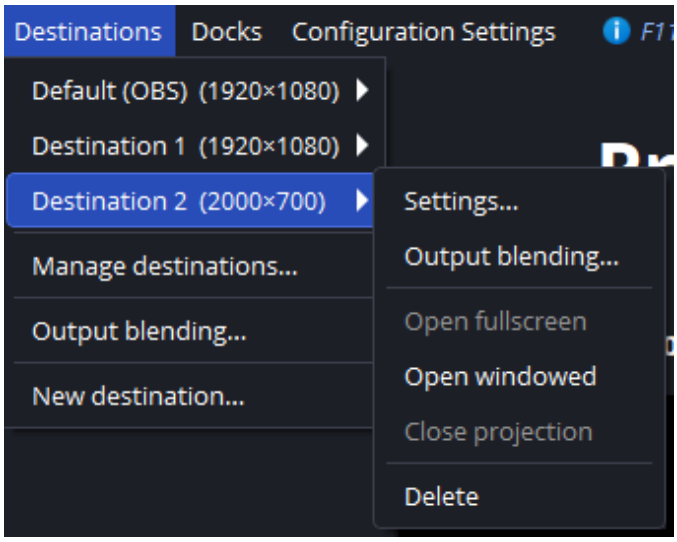


Figure 6. The Destinations menu with a destination submenu open.

Managing all destinations

Destinations → Manage destinations… shows every destination in one window. Select a destination on the left to change its name, resolution, output screen and lock, and to tick or untick the scenes it appears in; the window shows how many scenes cover it. Click **Apply** to save the changes; unticking a scene asks for confirmation, because the layout of that scene on this destination is deleted. The same window gives direct access to Output blending, fullscreen projection, Delete (Deactivate or Activate for the default destination) and New destination.

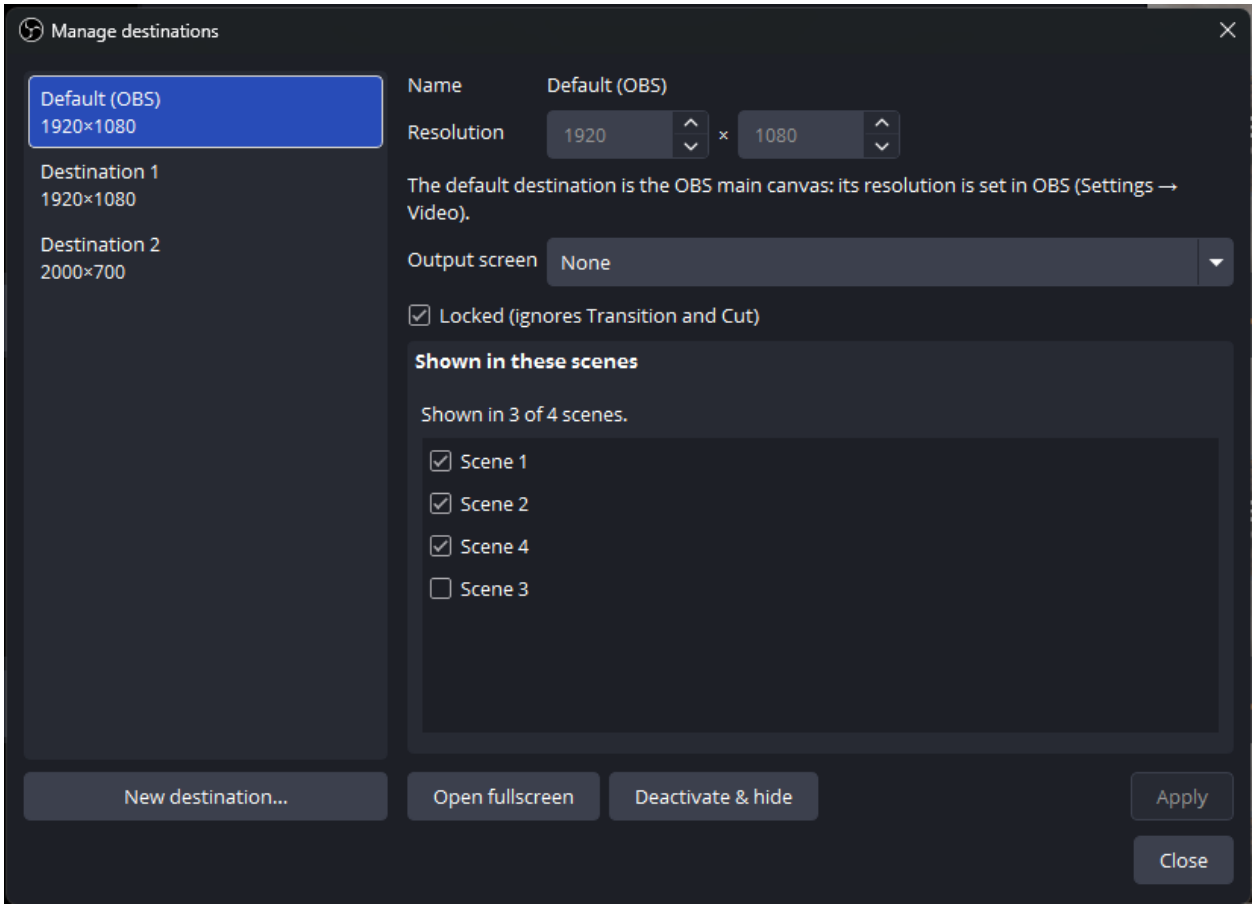


Figure 7. The Manage destinations window.

The default destination

The default destination is the OBS main canvas: its resolution is set in OBS Settings → Video, and it is what OBS streams and records. If you do not need it, choose **Deactivate & hide** in its submenu: it disappears from the studio, its scenes are kept for later, and streaming and recording are disabled until you reactivate it.

6. Scenes

Scenes are shared by all destinations. A scene decides **which destinations it covers**; each covered destination has its own version of the scene with its own sources.

Creating a scene

Click **+** in the Scenes dock. Give the scene a name and tick the destinations under **Show this scene on** (all are ticked by default). Scenes created directly in OBS, by a Stream Deck or through obs-websocket open the same dialog.

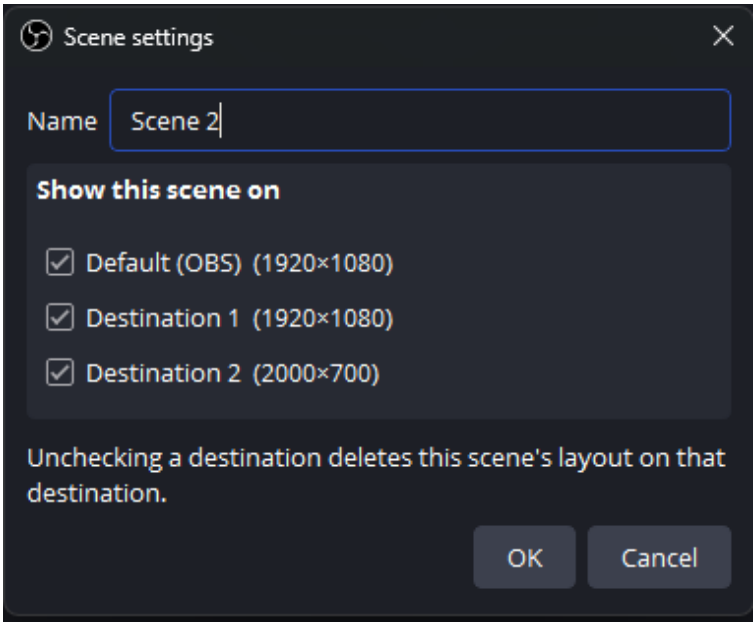


Figure 8. The scene dialog with its list of destinations.

Changing which destinations a scene covers

Double-click a scene, or use the gear button, to open the same dialog. Unticking a destination deletes the layout of that scene on that destination, so you are asked to confirm.

Organising scenes

- **Order**: drag a selected scene up or down, use the ↑ and ↓ buttons, or right-click for Move up, Move down, Move to top and Move to bottom. The order is shared with the OBS scene list.
- **Multiple selection**: click and drag across several scenes, or use Shift and Ctrl.
- **Remove**: press **Delete** or use the **−** button; a confirmation is always shown.

When a scene does not cover a destination

By default every destination stays visible. When the selected scene does not cover a destination, its Preview tile is hatched, framed in red and marked **Not in this scene**; its Program keeps showing what is live. You can instead show only the destinations of the selected scene (chapter 11).

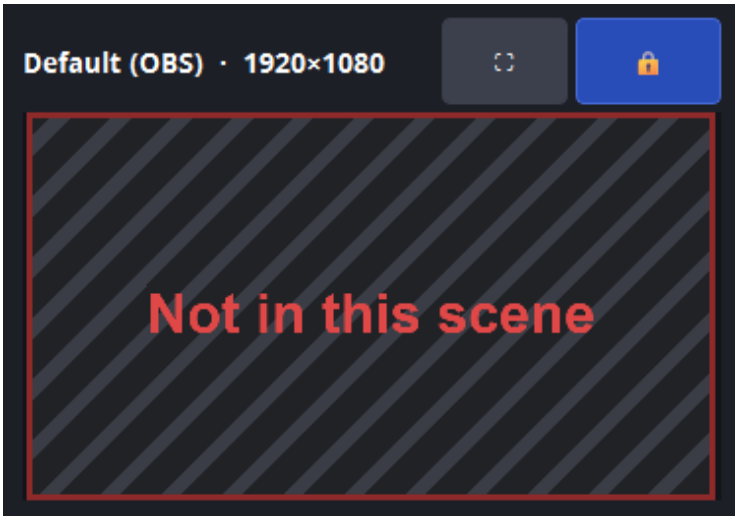


Figure 9. A destination that the selected scene does not cover.

7. Sources and editing

Click a Preview tile to make it the **active tile**: the Sources dock then lists the sources of that destination for the selected scene, and a discreet green frame marks the tile.

Adding sources

- **+** in the Sources dock lists every OBS source type, plus existing sources and other scenes.
- Drag image or video files from your computer onto a destination tile. They are added at their original size, centred; a new video shows its first frame without playing.

Editing with the mouse

Action	How
Move	Drag the source. It snaps to the centre of the canvas; hold Ctrl to disable snapping.
Resize	Drag a corner or edge handle; the aspect ratio is kept. Hold Shift to resize freely.
Crop	Hold Alt and drag a handle.
Nudge	Arrow keys move the selected source by one pixel.
Select	Click a source in the tile or in the Sources dock. Several sources can be selected in the dock by click-and-drag, Shift or Ctrl.
Reorder	Drag sources in the Sources dock, or use ↑ / ↓ and the Edit → Order commands.
Show / hide	Tick or untick a source in the Sources dock.

Selected sources have a green outline with handles; a source that goes beyond the canvas is outlined in orange. Clicking elsewhere in the window clears the outlines. A video that is not playing keeps a frame at its real size and can be edited like any other source.



Figure 10. Editing a source in Preview: selection handles and centre guides.

Edit menu and shortcuts

Command	Shortcut
Undo	Ctrl+Z
Copy / Paste reference	Ctrl+C / Ctrl+V
Remove (with confirmation)	Delete or Backspace
Copy / Paste transform	Ctrl+Shift+C / Ctrl+Shift+V
Reset transform	Ctrl+R
Fit to screen / Stretch to screen / Center to screen	Ctrl+F / Ctrl+S / Ctrl+D
Move up / down / to top / to bottom	Ctrl+↑ / Ctrl+↓ / Ctrl+Home / Ctrl+End
Properties, Filters	Edit menu, double-click or right-click

On macOS, use Cmd instead of Ctrl. Editing shortcuts work when the Preview or the Sources dock has the focus, so they never interfere with typing in other fields.

8. Going live

Transition and Cut

Select a scene: it appears in Preview on every destination it covers. Click **Transition** to send it to Program with the OBS transition and duration chosen in the Scene transitions dock, or **Cut** to switch instantly. All destinations switch at the same moment.

- A **locked** destination keeps its Program.
- A destination the scene does not cover keeps its Program.
- After the transition, the previous Program returns to Preview, as in OBS Studio Mode. You can choose to keep the scene you just took in Preview instead (chapter 11).
- Hotkeys for **Transition**, **Cut** and opening the studio can be set in OBS Settings → Hotkeys (they start with "Multidestination Studio").

Switching scenes in the normal OBS window only changes the default destination; the other destinations are driven from Multidestination Studio.

Editing what is live: Modify PGM

What is on air is a copy of the scene, so changes you make in Preview never reach the live output by accident. When you do need to fix something live, click **Modify PGM**: the button turns red and the Program tiles become editable with the same tools as Preview. Your changes are applied live and also saved into the scene. Click the button again to protect the Program.

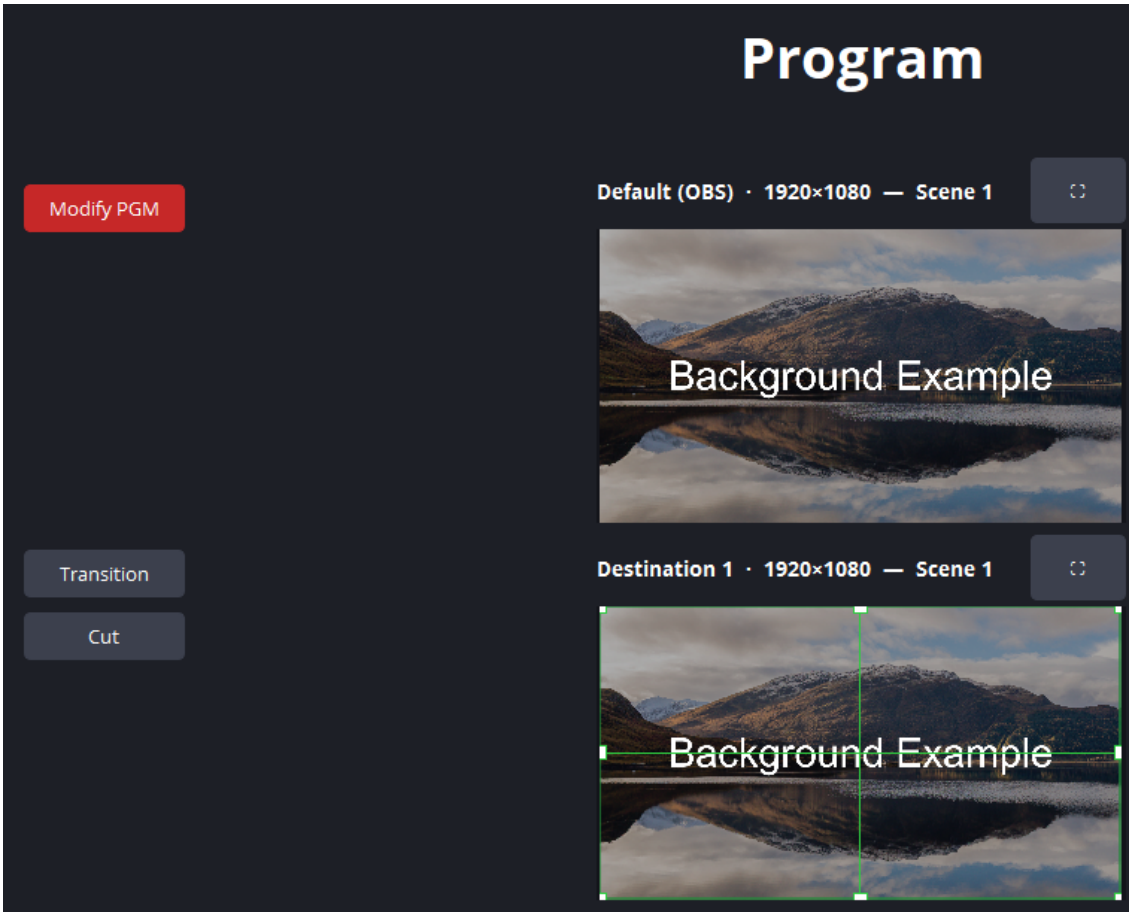


Figure 11. Modify PGM switched on: the button is red and the Program can be edited.

9. Output blending

Output blending spreads one destination across several outputs, typically two or more projectors whose images overlap on a wide screen. In the overlap, each projector fades out softly so the audience sees one seamless picture.

Creating a blended destination

1. Choose **Destinations → Output blending…**.
2. Click **New output** for each projector and choose its **screen**.
3. Drag each output from the list onto the **layout** (or double-click it). An output keeps the shape of its screen and cannot leave the layout.
4. Set the **layout size**, for example 3500 × 1080 for two overlapping full-HD projectors. The outputs follow automatically. The layout size becomes the resolution of the destination.
5. Place the outputs so that they overlap; overlaps are hatched and blended automatically.
6. Give the destination a name ("Destination X" if left empty) and click **Save & Create destination**.

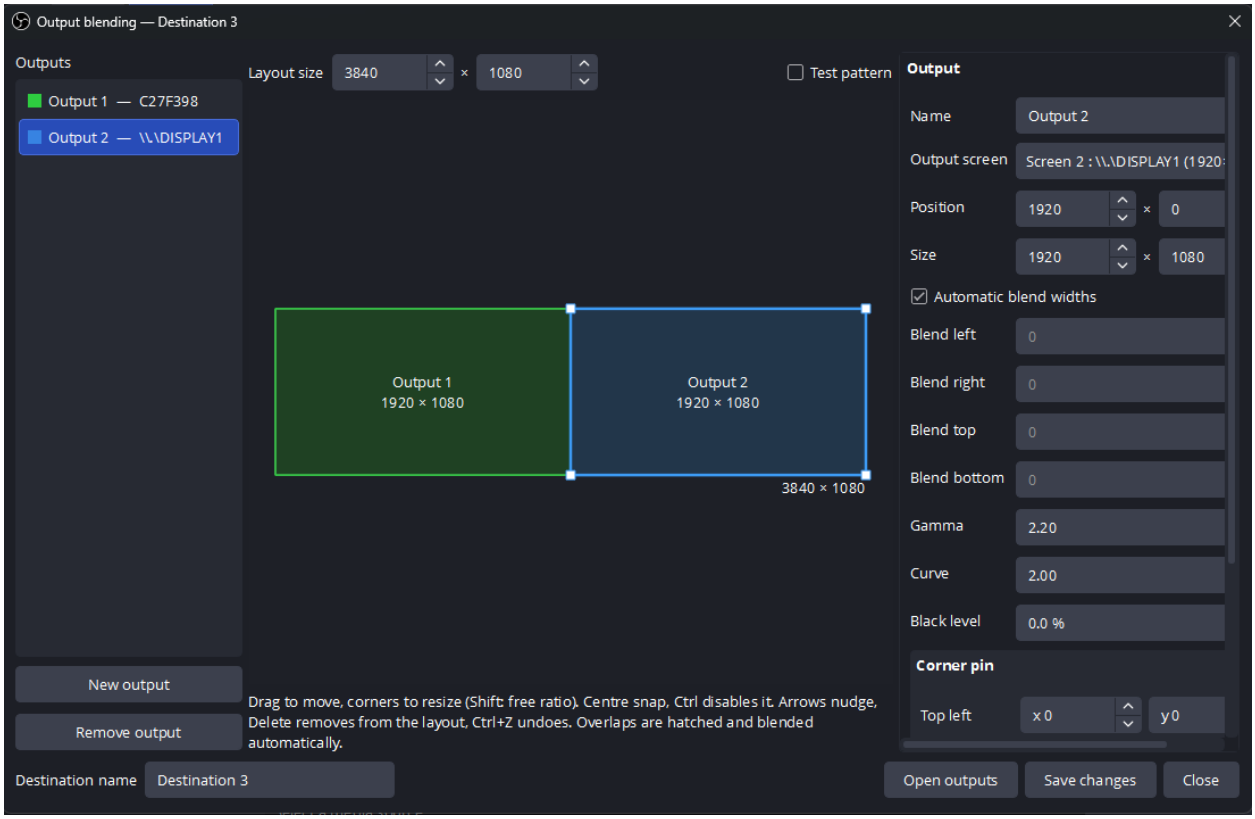


Figure 12. The Output blending window with two overlapping outputs.

Placing outputs

Outputs are moved and resized with the mouse like sources: corners resize while keeping the shape (Shift resizes freely), the centre snaps (Ctrl disables it), arrow keys nudge, Delete removes an output from the layout and Ctrl+Z undoes. Position and size can also be typed in.

Blend settings

Setting	Use
Automatic blend widths	On: the blend zones follow the overlaps. Off: set Blend left, right, top and bottom by hand, in layout pixels.
Gamma	Matches the response of the projector so the blend is even (usually 1.8 to 2.4).
Curve	Shape of the fade: 1 is linear, higher values are softer in the middle.
Black level	Raises the black outside the overlaps, so the overlap does not look brighter on dark images.
Corner pin	Moves each corner of the output, in output pixels, to correct keystone. Reset corners restores them.
Test pattern	Replaces the picture with a grid and centre cross to align the projectors.

Adjusting live and saving

For an existing blended destination, open **Output blending…** from its submenu or from its Settings. **Open outputs** shows every output full screen on its projector, and every change appears there immediately. **Save changes** stores the settings (the button greys out once saved). If you **Close** with unsaved changes, you are asked whether to **Save changes** or **Discard changes**; discarding returns the projectors to the saved settings.

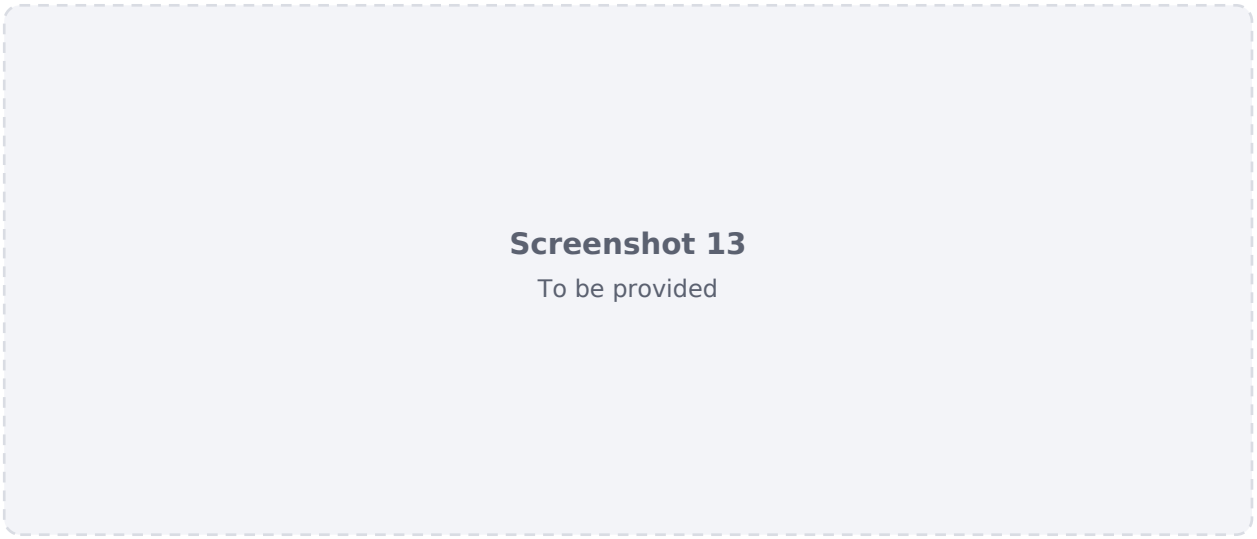


Figure 13. Two projectors showing the test pattern, aligned and blended.

> **Tip.** Start with the test pattern: align the projectors with the grid and corner pin, adjust the overlap, then the gamma, and finally the black level on a dark image.

10. Audio and media

The **Audio mixer** dock shows a strip per audio source with a level meter, a fader, a mute button and a monitoring menu (Monitor off, Monitor only, Monitor and output). It lists the active audio sources and those of the selected scene; **View all** shows every audio source. Audio is the OBS audio: it is mixed and sent once, with the default destination.

The **Media controls** dock controls the selected media source: play / pause, restart and stop, with its time.

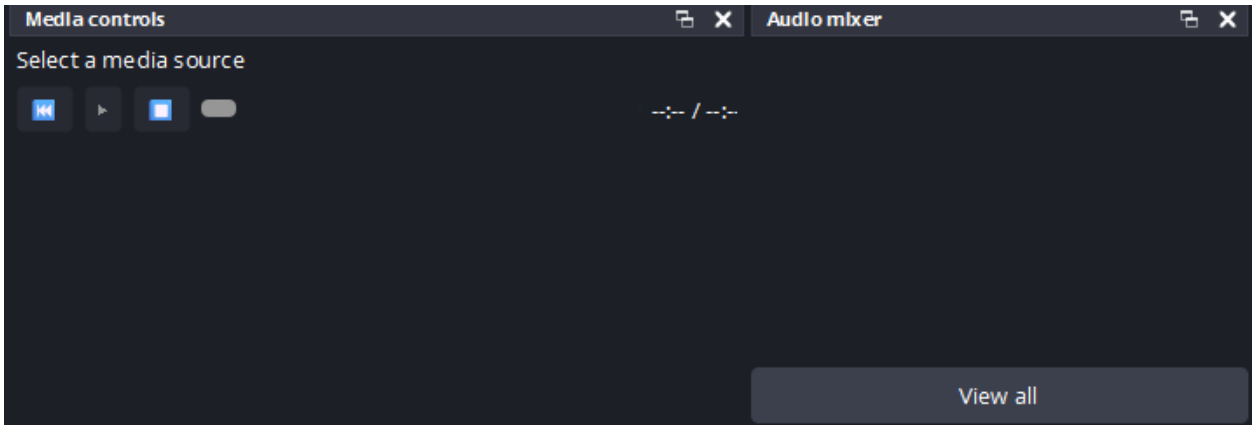


Figure 14. The Media controls and Audio mixer docks.

11. Configuration Settings

These settings apply to all your scene collections.

Setting	Choices
After a transition…	**The previous Program returns to Preview** (recommended, default), or **The scene sent live stays in Preview**, useful when your scenes do not cover the same destinations.
Destinations shown…	**All destinations, all the time** (recommended, default), or **Only the destinations of the selected scene**.
Startup Settings…	**Launch Multidestination Studio on startup**: on or off (off by default).
Language…	Same as OBS, English, Français or Norsk (Bokmål). Takes effect when OBS restarts.

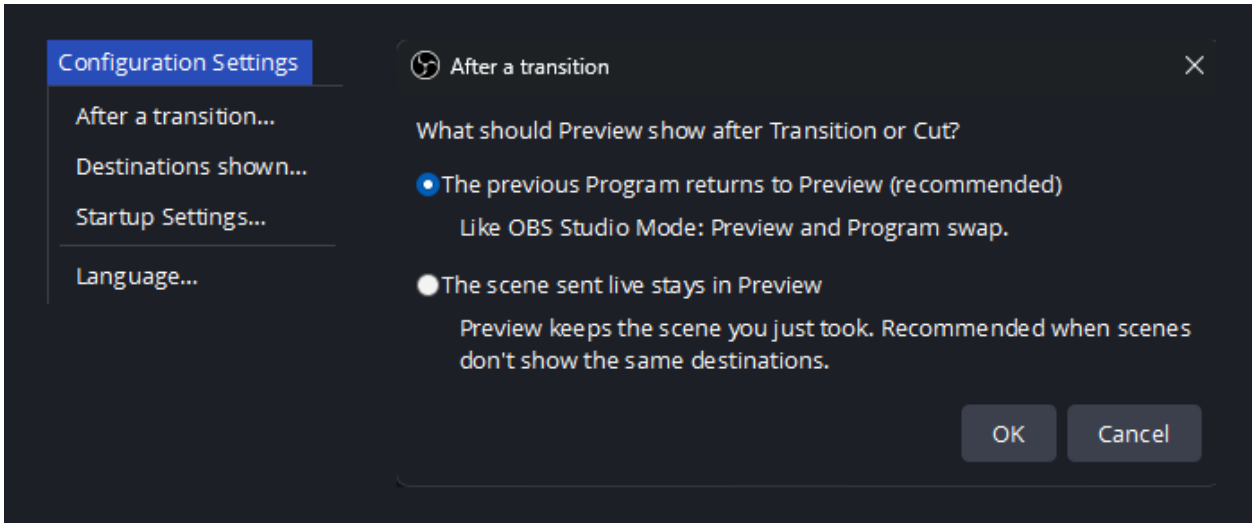


Figure 15. The Configuration Settings menu and the After a transition dialog.

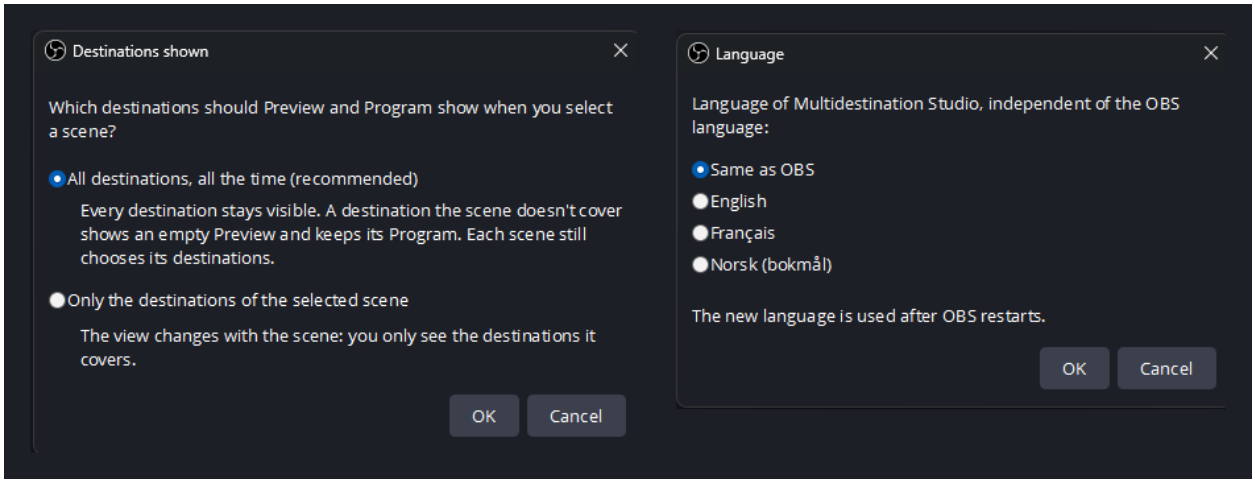


Figure 16. The Destinations shown and Language dialogs.

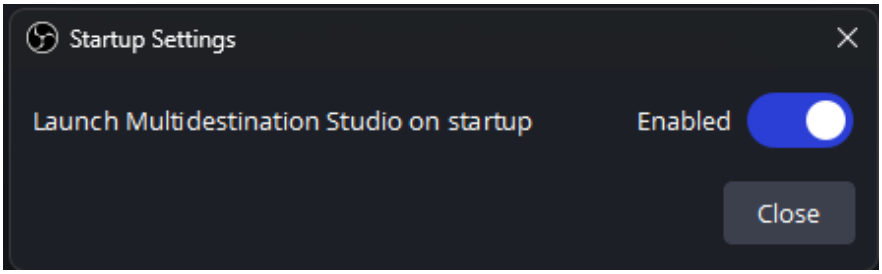


Figure 17. The Startup Settings dialog.

12. Saving, shortcuts and troubleshooting

What is saved, and where

- Destinations, their scenes and layouts, output blending and the dock arrangement are saved in the OBS **scene collection**. Switching collection switches all of them.
- Configuration Settings are saved in your OBS user settings and apply to every collection.

Shortcuts at a glance

Key	Action
F11	Full screen on / off
Esc	Leave the studio; in a projection window, close the projection
Delete	Remove the selected scenes or sources (with confirmation)
Ctrl+Z	Undo
Arrow keys	Nudge the selected source or output
Shift (while resizing)	Free aspect ratio
Ctrl (while moving)	Disable centre snapping
Alt (on a handle)	Crop

Troubleshooting

Problem	What to check
The button does not appear in OBS	Check that OBS is 31.1 or later and reinstall with OBS closed. In OBS, Help → Log Files → View Current Log should contain lines starting with [multidestination-studio].
Windows blocks the installer	The installer is not code-signed. In SmartScreen choose More info → Run anyway; in Defender, allow the file from the protection history.
A destination stays empty	Check that the selected scene covers it and that it is not locked.
Open fullscreen is greyed out	Choose an output screen in the destination Settings, and check that the screen is connected.
The interface is in the wrong language	Configuration Settings → Language, then restart OBS.